

An Introduction

Course Mathematical Logic

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about the course

Course Name:	Mathematical Logic	
Field:	ICT	
Teaching hours:	30 hrs	
	Lecture	18 hrs
	Exercises	12 hrs
	Practicals	06 hrs
Self-Study hours:	36 hrs	

about the lecturer & communication

Name: **Thanh-Sach LE**
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Communication: Put **[MATH.LOGIC-2014]** on the subject of
every email for this course

- 1 **Mathematical Logic** a subject that provides capacity of **modeling**, **proof**, and automatically **reasoning** to many key areas in computer science, for examples, Artificial Intelligence, Software Engineering, Programming Language, and so on.
- 2 This course is to provide to students a solid introduction to **Propositional Logic**, **Predicate Logic**, and principles to perform proof and reasoning in those forms of logics. Moreover, it also introduces to students **libraries** and **algorithms** to develop automatically proof and reasoning computer systems.

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Course's Objective

By the end of the course, successfully completed students will:

- be able **to explain** and **to give** examples on how to write syntax-correct sentences in Propositional Logic and Predicate Logic, on how to interpret the their semantics, and how to do reasoning/proving with Natural Deduction, Resolution, and Hilbert Systems(Level: 2*)
- be able **to model** problems in real world using Propositional Logic or Predicate Logic, and then **to solve** the problems formally (Level: 3,4)
- be able **to compare** expressional capacity and reasoning methods between Propositional Logic or Predicate Logic (Level: 4*)
- be able **to create** computer programs that are able to do automatically reasoning using Propositional Logic or Predicate Logic, using open-source libraries (Level: 5*)

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See attached file.

- ① Patrick J. Hurley, "**A Concise Introduction to Logic**, 7th Edition, Wadsworth Publishing, 2000.
- ② Shawn Hedman, **A First Course in Logic: An Introduction to Model Theory, Proof Theory, Computability, and Complexity**, Oxford University Press, 2004.
- ③ Anita Wasilewska, **An introduction to classical and non-classical logics**
- ④ Stuart Russell and Peter Norvig, **Artificial Intelligence: A Modern Approach**, 2nd, Prentice Hall, 2003.
- ⑤ Miklos ferenczi Miklos szots, **Mathematical Logic for applications**, 2011

Grading Policy

total: 100%

Component	%
Attendance/Attitude	10%
Exercise(s)	20%
Practicals	10%
Mid-term test	20%
Final exam	40%